Party Above All: The Determinants of Emotional Tone in French Campaign Manifestos

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Abstract

This paper examines the determinants of emotional tone in French legislative campaign manifestos using a corpus of 6,446 texts from the 1981 and 1993 elections. It combines linguistic analysis, topic modelling, emotional measurement in texts and pictures to show that party affiliation is the main driver of emotional tone. By contrast, local economic conditions and incumbency status have limited effects. A comparison between text and images further reveals that candidates present neutral or positive facial expressions regardless of the tone of their manifestos.

1 Introduction

Emotions play an important role in politics. Even though voters are often seen as rational [1], research shows that feelings influence how people form opinions, whether they vote, and who they vote for. Thus, politicians and candidates might be inclined to use emotions strategically. This is evidenced by the increasing use of violent and populist rhetoric to connect with citizens at an emotional level [2]. Grounded in retrospective voting theory, scholarship shows that the emotions candidates deploy in their campaigns depend on their incumbency status, their policy position and onngoing events [3].

Historically, the scholarship of emotional appeals by politicians has been dominated by the study of American elections [4], but a growing body of evidence has shifted attention toward cross-national contexts [5], including comparative studies across European party systems [3]. Yet little evidence has focused on specific national contexts outside the USA. In this paper, I investigate the determinants of emotional tone in the French political context, using manifestos from the legislative elections of 1981 and 1993.

1.1 Data and Context

The primary data source for this study is the Archelec archive, a digitized collection of French electoral manifestos covering presidential and legislative elections from 1958 to 2012, hosted on Internet Archive and maintained by the CEVIPOF (Sciences Po) [6].

I focus on two legislative elections: 1981 and 1993. The 1981 election brought the left to power, with the Socialist Party winning a large majority after its member François Mitterrand's presidential historical victory. In contrast, the 1993 legislative election saw a strong victory for the right, with the *Rassemblement Pour la République–Union pour la Démocratie Française* coalition winning a large majority, leading to a period of cohabitation with the left. I focus on these elections because they reflect different political configurations (in terms of who is in power) and socio-economic contexts, both of which are likely to shape the emotional tone of party manifestos. While 1981 occurred in the wake of a historic left-wing presidential win, 1993 followed a period of economic recession [7] .

*This project was conducted alone as part of the beginner NLP track — https://github.com/selaazdoudi/manifesto-emotional-tone

1.2 Natural Language Processing Problem Formulation

The dataset used in this study includes 6,446 party manifestos, with an average length of around 700 words per document, adding up to approximately 4.5 million words in total. As Justin Grimmer and Brandon M. Stewart (2014) point out, the massive growth of political documents makes it difficult to rely only on human coding for computational political science [8]. Thus, a series of NLP techniques are used here for the measurement of discrete emotions across parties and elections.

1.3 Literature review

1.3.1 Determinants of Emotional Tone in Elections

Scholars generally point to three main explanations for variation in emotional campaigning: party identity, access to power, and changing external conditions.

Ideology and Party Identity. One explanation has to do with party identity. Research on so-called outsider parties shows that they often need to stand out clearly from mainstream competitors [9]. One way they do this is through emotional messaging. The literature shows that parties on the political extremes are more likely to rely on strong negative emotions to grab attention and energize their supporters. For instance, Marcus Nai and Jürgen Maier, in a large cross-national study of elections between 2016 and 2020, find that more extreme candidates tend to use emotions like fear more often than positive or reassuring tones [10].

Power and Status Dynamics. Another explanation focuses on whether a candidate is in power or trying to win it. Candidates competing from an advantageous position (most notably, incumbents) have incentives to build enthusiasm among the electorate, whereas those in a less favorable position need to persuade more voters to support them. In this context, anxiety appeals can make voters switch sides by weakening existing party ties.

Candidates competing with a comparative advantage, therefore, tend to adopt a more positive and optimistic tone. Nai and Maier show that incumbents and frontrunners rely more on enthusiastic messaging in their campaigns, but do not find evidence for the inverse mechanism for fear appeals [10]. Similarly, Travis Ridout and Kathleen Searles find that incumbents are less likely to rely on anger in the context of American Senate elections [11].

Context and External Events. A third explanation looks at the broader context. Events like economic crises, wars, terrorism, or health emergencies might shape politicians' use of emotions. For example, Kym De Castella, Craig McGarty, and Liam Musgrove studied speeches by Australia's Prime Minister John Howard (1996–2007). They found that fear was common, especially before the Iraq War and when government support was weakening [12].

Economic conditions also play a role. Based on an analysis of 400 party manifestos across eight European countries, Crabtree et al. (2020) find that parties use less positive language when the economy is performing poorly, particularly in terms of inflation and unemployment [3].

1.3.2 Prior NLP Approaches to Emotion Measurement

Most approaches rely mainly on "bag-of-words" methods (lexicon based methods), which count the presence of certain words [12]. While simple, these methods have important limitations: they are often designed for specific contexts and cannot take into account the meaning of words in context. To address this, more recent methods use word embeddings, which learn the meaning of words based on how they appear with other words. However, these approaches still assign a fixed meaning to each word, regardless of the sentence in which it appears. Newer models based on deep learning, such as BERT, improve on this by taking context and order into account [13].

A key challenge in this literature is how to validate emotion measures. In practice, machine learning methods need annotated datasets to check how accurate they are. But creating these datasets is costly and time-consuming, which is why researchers often rely on bag-of-words approaches, even though their limitations are well known [12].

Birkenmaier et al. (2024) distinguish between internal and external ways to validate results. The strongest external approach is to compare automated scores to a fully hand-coded "gold standard"

dataset, but this is costly and difficult to produce [15]. Because of this, researchers often use simpler alternatives. One is crowdcoding, where a sample of texts is rated by several non-experts online. This is cheaper and can still be reliable—Haselmayer and Jenny (2017) show that crowd ratings can be close to expert ones [16]. Internal validation methods are also used. It is recommended to check some texts by hand (face validity) and to compare results across tools (convergent validity) to see if classifications make sense [15].

1.4 Research Question Framing

This paper asks: what are the determinants of the emotional tone of French legislative campaign manifestos? It focuses on four questions:

- Do candidates at the ideological extremes use more emotionally intense language than mainstream candidates?
- Does local economic context, measured by unemployment, influence the emotional tone of manifestos?
- Do incumbents and candidates from governing parties use more positive rhetoric than challengers and opposition candidates?
- Do facial expressions in campaign photographs align with the emotional tone of written manifestos?

2 Data

2.1 Corpus Construction

Each text file corresponds to a single candidacy (with a candidate and their substitute in a single constituency). The raw corpus is composed **8,958 campaign manifestos** (3,182 from the 1981 election and 5,936 from 1993).[2]. Candidate metadata (name, party, party affiliation, gender, age, profession, electoral district) is drawn from the same archive and merged on a unique document identifier. Only 1.75% of manifestos could not be matched to a candidate, and these unmatched documents are dropped from the analysis.

2.2 Political Classification

Classifying candidates ideologically is not straightforward: there were 270 party labels in 1981 and 307 in 1993, making a fully manual approach hard to implement.

Instead, I match each party label to an official nuance code from the Centre de Données Socio-Politiques (CDSP) open dataset. This is done in three steps:

- A dictionary match using the CDSP nuance fields;
- Manual fixes for 639 remaining cases;
- Dropping regionalist and unaffiliated candidates, who cannot be clearly placed on the left–right scale.

This results in **6,446 manifestos** across the political groups shown in the figure below.

[2]The Archelec archive does not claim to be exhaustive: as its curators acknowledge "exhaustivity is a myth", since electoral manifestos were never subject to systematic collection [6].

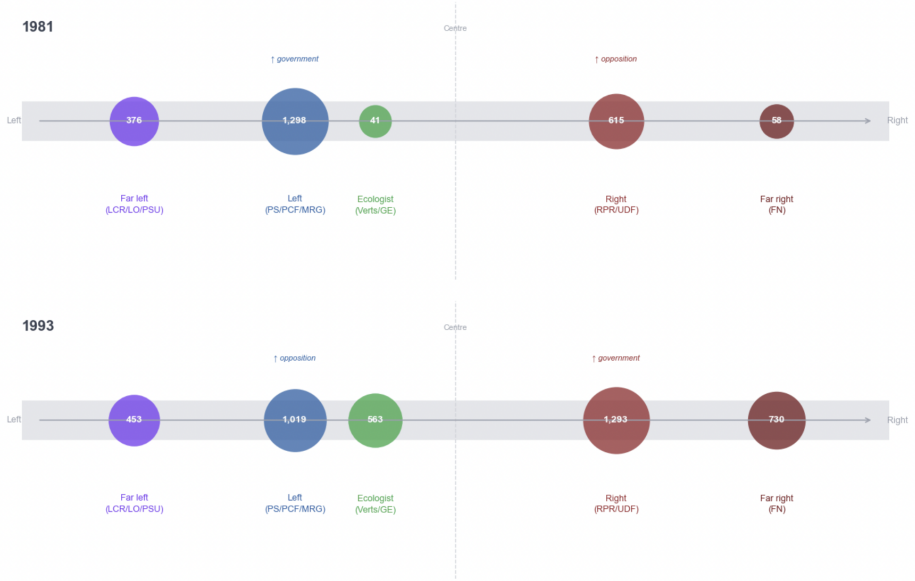

Figure 1: Ideological spectrum: candidate counts by political group

2.3 Contextual Variables

Office Holding This variable captures whether a candidate held any elected office at the time of the election, based on one of the available fields in the raw metadata. Candidates with a current mandate—whether local, regional, national, or European (e.g. mayor, councillor, minister, Member of the European Parliament) are coded as 1, and those without one as 0. Overall, 54.6% of candidates in the analytical sample held an office at the time of the election, with a stable rate across both elections (54.2% in 1981 and 54.9% in 1993).

Departmental unemployment rate This variable is drawn from INSEE data (`https://www.insee.fr/fr/statistiques/1409932`). Following the literature [11], I use lagged values (1992 for the 1993 election) to reflect the economic context at the time candidates wrote their manifestos. As the data is not available before 1982, this variable is only used for the 1993 election, which is the most relevant election given the economic context mentioned above.

2.4 Face Extraction

Images are downloaded from the URLs in the metadata. Faces are then detected using an adapted library, cropped, and saved as individual images. Many URLs were broken, so I recovered 3,518 faces from the manifestos, and some manifestos did not contain any faces to begin with.

3 Methodology

3.1 Text-Based Emotion Measurement Strategy

I use the six basic emotions identified by Paul Ekman : joy, sadness, fear, anger, surprise, and disgust, typically used as a benchmark in the study of emotion [22]. This goes beyond the more common sentiment analysis in political science, which reduces content to positive or negative [3].

3.1.1 Method 1 : Lexicon-Based Scoring (FEEL)

The first method uses FEEL by Abdaoui et al. (2017). It is a French version of the widely known (and used) NRC Emotion Lexicon, translated from english, expanded with synonyms, and validated by a human translator. It contains over 14,000 French words linked to positive or negative sentiment and to Ekman's emotions described above. Tests on French data, including **parliamentary debates**,

show that FEEL performs well for sentiment analysis and outperforms other French lexicons for emotion classification [17].

This method operates at the **word level**. Each text is tokenised, and words are matched to the lexicon. Emotion scores are computed by aggregating these matches across the entire document. As noted in the literature review, FEEL is a bag-of-words method and therefore does not consider word order or context.

3.1.2 Method 2 : Fine-Tuned Transformer (DistilBERT)

I use a DistilBERT model fine-tuned on about 200,000 US political texts (tweets, campaign speeches, and congressional speeches) annotated with GPT-4o-mini. The model predicts eight emotion scores between 0 and 1: anger, sadness, fear, disgust, pride, joy, gratitude, and hope, from which I retain Ekman's emotions for better comparison [18].

Because the model is trained on English, each manifesto is first translated from French using *facebook/nllb-200-1.3B*. To preserve the full text (due to the model's input limit), it is split into segments before translation and then recombined [19].

For emotion scoring, the translated text is divided into chunks of about 500 tokens, due to the model's input limit. Each chunk is processed separately, and the final score is the average across all chunks.

The **model therefore works on sequences of words** and takes into account word order and local context, as opposed to FEEL.

3.1.3 Method 3 : Zero-Shot Classification (mDeBERTa)

The third method uses a multilingual natural language inference (NLI) model applied directly to the original French text, so no translation is needed. The model is not trained specifically for emotion detection. Instead, it tests statements such as *"This text expresses anger"* and estimates how well the text supports each one. To cover the full manifesto (because of the model's input limit), each text is split into chunks of about 500 tokens. Each chunk is scored separately for anger, fear, sadness, disgust, and joy, and the final score is the average across all chunks. As the DistilBERT model, this **approach works on sequences of words and takes context and order into account given the input**.

3.2 Text-Based Emotional Measurement Validation

3.2.1 Convergent validity

Following Birkenmaier et al. (2024) and Van Atteveldt et al. (2021), I compare the three methods on the same texts. All three assign the same dominant emotion in only 3.8% of cases, while at least two agree in 49.5%.

Agreement is higher for valence (positive vs. negative sentiment): all three models agree in 11% of texts. Pairwise agreement is uneven. FEEL and DistilBERT agree in 77.5% of cases, compared to 33.3% for DistilBERT and the zero-shot model, and 11.2% for FEEL and the zero-shot model. These figures begin to suggest that the models capture different aspects of the texts.

To understand these differences, I conduct face validity checks. I manually review texts where all three models disagree and assess whether the assigned emotions match the content. I then test whether the same patterns hold across the corpus using unsupervised machine learning techniques.

3.2.2 Face validity

I read 50 manifestos on which the three models disagree and compare their **dominant emotion** labels with my own interpretation. These texts come from the 3,254 cases (50.5%) where each model assigns a different dominant emotion. Three patterns emerge.

DistilBERT tracks best emotional tone. It identifies joy in celebratory post-Mitterrand manifestos and anger in combative texts. Jacques Roux (PCF) opens with "*c'est l'espoir qui habite le cœur de chacun*" (joy 0.62). In combative texts, Jo Gallet (PCF) asks "*pourquoi faudrait-il accepter que ça aille toujours plus mal ?*" (anger 0.56), and a Lutte Ouvrière manifesto urges "*nous devons nous fâcher*" (anger 0.67).

The NLI model conflates topic and tone. mDeBERTa assigns sadness in 43 of 50 cases, including clearly celebratory texts such as Roux's (PS) "*c'est l'espoir...*" (0.95) and Notebart's (PS) "*la victoire vous appartient*" (0.99), as well as combative texts like the Lutte Ouvrière manifesto ("*nous devons nous fâcher*"; 0.98). Labels seem to follow words such as *chômage* or *pauvreté* rather than the tone of the text. This reflects a limitation of NLI-based zero-shot classification: the model confuses "what is this text about?" rather than "what is its emotional tone?". Therefore, emotional classification seems to fall outside of the general tasks zero-shot models are designed for.

FEEL assigns anger or fear and never identifies joy. Positive texts such as Aznar's (Entente des Écologistes) "*un souffle d'air dans la vie politique*" (anger 0.11) are classified as anger, while combative texts are inconsistently labelled (the Lutte Ouvrière manifesto cite above is classified as fear 0.18).

3.2.3 Corpus-level validation

DistilBERT tracks emotional tone more accurately than the other two models A clear example comes from left-wing manifestos written in June 1981, just after the victory of François Mitterrand. These 1,298 texts are about the same political moment and share a similar tone: they are largely celebratory. DistilBERT reflects this, assigning joy in 86% of cases. The zero-shot model does not. It assigns joy in only 5% of texts, and instead classifies 47% as sadness and 46% as anger. The following tests help explain this difference.

Test 1. The zero-shot model conflates topic and tone. To examine this more closely, I turn to the 1993 legislative elections, when unemployment was the main issue during a period of severe recession, with rates reaching 11.5%. This is a useful case because most manifestos focus on the same topic. Any differences in predicted emotions should therefore come from how candidates talk about unemployment, rather than from the topic itself.

I use BERTopic to group texts that discuss the issue in different ways. BERT works well here because it understands words in context, so it can pick up differences in tone and framing. It identifies four clusters around employment:

- conflictual (*patron, travailleur, salaire*)
- programmatic (*partage du travail, école, ouvrier*)
- principled (*droit, égalité, lutte*)
- mobilising (*privatisation, licenciement, unité*)

These clusters show that candidates frame the same issue differently. If the zero-shot model captured emotional tone, its predictions should vary across these clusters. Instead, it assigns sadness to almost all texts in every cluster, between 99.6% and 100%, with almost no variation (std = 0.02, 0.05, 0.12, 0.14).

Test 2. FEEL systematically misses positive emotions. FEEL shows a different limitation. It assigns joy in only 52 of 6,446 texts, or 0.8%. This likely happens because manifestos almost always include descriptions of problems, which the lexicon detects even when the overall tone of the text is positive.

3.2.4 Model selection

DistilBERT provides the most reliable measure of emotional tone. It aligns with manual readings and remains sensitive to rhetorical register at corpus level. By contrast, mDeBERTa reflects topical content rather than tone, and FEEL is heavily skewed toward negative emotions. I therefore rely on DistilBERT in the main analyses.

3.3 Image-Based Emotion Measurement Strategy

For image-based emotion analysis, I use the *facial_emotions_image_detection* model, which is proven to achieve about 91% accuracy [23].

4 Results

4.1 Emotional Intensity Across the Ideological Spectrum

Based on the literature review above, I expect candidates at the ideological extremes to use more emotionally intense language than mainstream candidates. This would produce a U-shaped pattern for negative emotions and the opposite pattern for positive emotions. Figure 3 supports hyothesis.

The U-shape is clearest for negative emotions, especially anger, although it is not perfectly symmetric. Candidates on the far left and far right consistently score higher than mainstream candidates, while ecologists, who sit closer to the centre, score the lowest. The pattern is strongest for anger (FL = 0.49, FR = 0.41, E = 0.14) and also appears, though in a slightly weaker way, for fear and sadness. The far left generally scores higher than the far right, suggesting that in the 1981 and the 1993 elections, the far left used substansively more emotional language than the far right. Joy shows the opposite trend. Ecologists score the highest (0.39), while candidates at both extremes score the lowest.

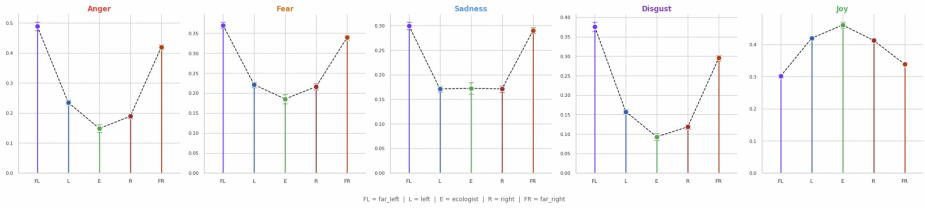

Figure 2: Emotional intensity across the ideological spectrum

As a robustness check, I compare emotional intensity (a composite of emotions) across the three models described in the methodology. The U-shaped pattern remains consistent (see Figure **??**).

4.2 Local Economic Context and Emotional Intensity

This section examines whether local unemployment shapes campaign rhetoric, in terms of both issue salience and emotional tone. It focuses on the 1993 legislative election, held in the aftermath of an economic recession.

4.2.1 Identifying Unemployment Discourse via LDA

I estimate a Latent Dirichlet Allocation model (LDA) with $K = 7$ topics on the 1993 manifesto corpus. The number of topics is chosen using a standard coherence and stability criterion, combining the c_v coherence score and Jaccard similarity across neighbouring values of K (see Figure 7).

I use LDA, as opposed to BERT in Section 3.2.3, since the goal here is not to capture differences in how texts are written, but to use word co-occurrence to identify texts that discuss unemployment.

Three topics relate to unemployment: Topic 1 links it to taxation and immigration, Topic 3 takes a more programmatic approach, and Topic 5 focuses on labour conflict. Topic 2 is not selected for the analysis because of low coherence.

4.2.2 Local Unemployment Does Not Drive Rhetorical Choices

As evidenced by the literature, the expectation is straightforward: if local conditions matter, candidates in high-unemployment areas should talk more about unemployment, and possibly do so in a more negative tone.

However, the data do not support this. First, the share of manifestos focused on employment is almost the same everywhere: 35.8% in low-unemployment areas, 34.3% in average areas, and 37.0% in high-unemployment areas. The difference is very small.

Second, within employment-focused manifestos, emotional scores change very little with local unemployment. The relationship is almost flat for anger, fear, disgust, and sadness. Scores are slightly higher in high-unemployment areas, but the differences are small and there is no clear trend as evidenced by the following figure.

Table 1: LDA Topic Structure — 1993 Manifesto Corpus ($K = 7$)

Topic	c_v	Top words	Interpretation
0	0.58	front_national, immigration, impôt, immigré, défense, famille	Discourse focused on immigration, national identity, and defence, with little emphasis on economic issues
1	0.85	impôt, immigration, chômage, emploi, retraite, travail	Links unemployment and immigration to taxation and welfare
2	0.33	emploi, immigration, alliance_populaire, chômage, entreprise	Mixed discourse combining economic and identity themes; low coherence suggests no clear structure
3	0.44	communiste, emploi, force, progrès, socialiste, service, chômage	Frames employment in terms of public services, rights, and political action
4	0.39	emploi, environnement, travail, écologie, maire	Mixed topic combining employment, environment, and local politics
5	0.70	travailleur, emploi, entreprise, patron, salaire	Focus on labour relations and conflict between workers and employers
6	0.61	homme, femme, conviction, centre, solution_concret	General discourse focused on everyday concerns and practical solutions

Note: c_v indicates topic coherence. Topics 1, 3, and 5 are used as the unemployment-related subset in the analysis.

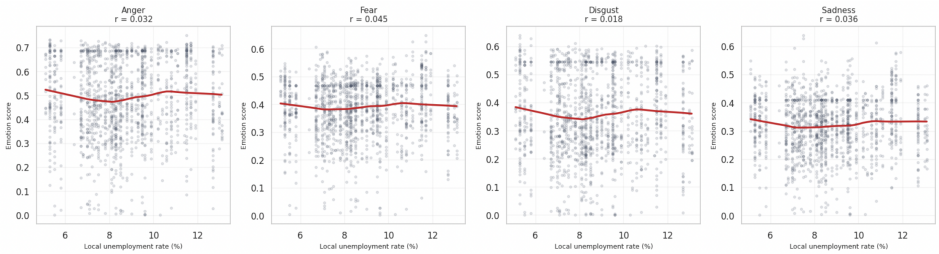

Figure 3: Local unemployment rate vs. emotional scores

4.2.3 Party Affiliation, Not Local Context, Drives Emotional Tone

Looking at the LDA topics more closely, they line up very clearly with political groups. When I compare topics to my political classification variable, the match is strong: Topics 0, 1, and 2 are almost all far right; Topic 5 is almost entirely far left (99.1%); Topic 3 is mostly institutional left (76.9%); and Topic 6 is entirely mainstream right (100%). What looks like different ways of talking is largely just a reflection of party affiliation, which goes in hand with the results from the following section.

4.3 Incumbency, Governing Party, and Positive Rhetoric

Legislative elections work at two levels. Locally, candidates run as incumbents or challengers. Nationally, they belong either to the party in power or to the opposition. These two do not always overlap: a candidate can be an incumbent but still be in the opposition.

This section looks at both. It asks whether incumbents use more positive language than challengers, and whether candidates from the governing party use more positive language than those in the opposition. The analysis focuses on the 1981 and 1993 elections. Emotional tone is measured using joy and a broader positivity score.

Based on the literature review above, candidates in a stronger position should sound more positive. Incumbents are expected to be more positive than challengers, and governing party candidates more positive than the opposition. The governing party effect should be stronger.

The results support this. Incumbents are slightly more positive than challengers. They score higher on both joy (0.43 vs. 0.39) and overall positivity (0.46 vs. 0.43), but the difference is small. The effect is much larger for the governing party. Candidates in power are clearly more positive than those in the opposition. The gap is about twice as large (and statistically significant) as the incumbency effect. This pattern holds in both elections, although it is stronger in 1981 than in 1993.

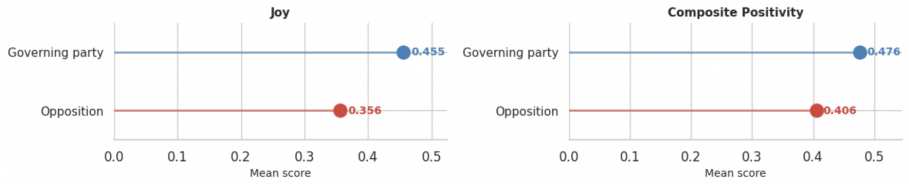

Figure 4: Positivity by Governing Party Status

Overall, party matters more than incumbency, which is aligned with the conclusions from the two previous research question.

4.4 Facial vs. Textual Emotion Signals

Beyond text-based emotion measurement, I complement the analysis with a facial emotion detection approach. For each manifesto, I extract the candidate's photograph and apply a facial emotion detection model to score emotions from the face. This allows me to compare the emotional signal conveyed by the candidate's visual presentation with the emotional tone of their written text.

The two signals do not align. Facial scores are systematically lower (and statistically significant) than text scores for negative emotions (anger, fear, disgust, and sadness, while joy is the only emotion where face and text converge (face = 0.399, text = 0.416). It is not surprising that joy is the most dominant emotion. This gap is expected: campaign photographs are a highly controlled communication medium, where candidates typically project a neutral or positive image regardless of the rhetorical tone of their text. Given that voters form rapid impressions from facial appearance, candidates have strong incentives to present favorable visual cues [24]. The face conveys the persona; the text conveys the message.

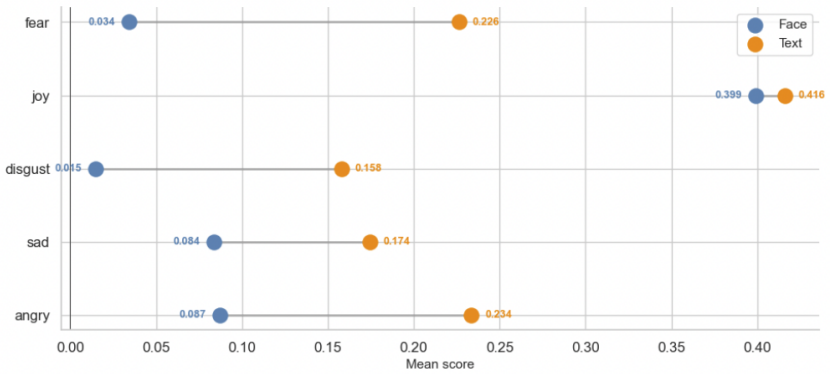

Figure 5: Face vs Text emotions — mean scores

5 Conclusion : Limitations and Future Directions

This study has several limitations. Although the emotion measurement method has been validated, it remains imperfect. It relies on a model trained on American political texts and applied to French

manifestos through translation, which may affect accuracy. No suitable BERT-type model trained on French political texts and emotions was available. A more appropriate approach would be to annotate a sample of French manifestos to create a domain-specific dataset, and then fine-tune a French-language transformer on this data.

The measure of incumbency is also approximate, as holding local office does not provide the same advantages as a national mandate. In addition, the facial emotion analysis is limited by data availability, with a reduced sample of images that may not be fully representative.

At the same time, the results are consistent across methods (LDA, BERT, etc.) and point to a clear conclusion: party affiliation dominates the other factors considered. This finding holds across different approaches and across two elections with different contexts. The study does not include individual characteristics such as gender or profession, but party affiliation shows a stronger and more consistent effect than both incumbency and local economic conditions, which already capture some individual and contextual variation, suggesting it would likely remain the dominant factor even if additional variables were included.

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A Figures

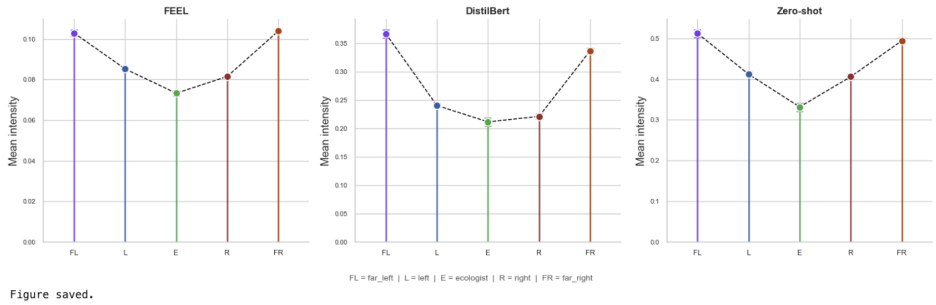

Figure 6: Mean Emotional Intensity by Political Block

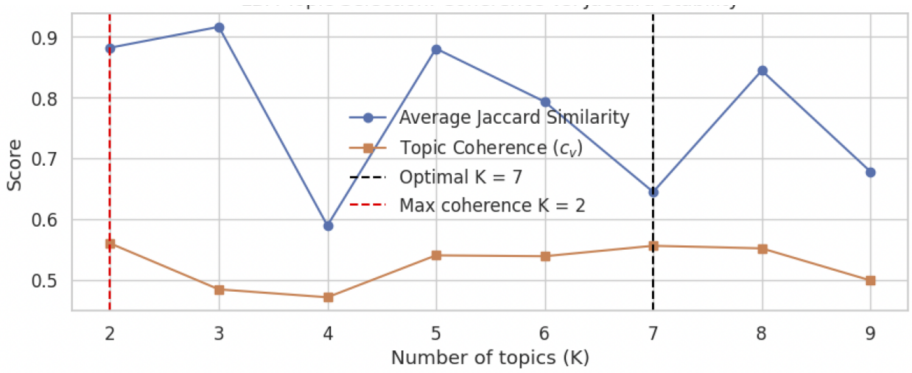

Figure 7: Topic Selection: Coherence and Stability Criteria